

GeoNews — Week 30/2026

A weekly digest on Quaternary palaeoclimate research, East African rift dynamics, palaeohydrology and time-series analysis · 20 July 2026

What is GeoNews? An **automatically generated weekly digest**. Every Monday a research workflow screens the publications of the past one to two weeks along with funding, policy and community sources, scores the hits against a fixed set of criteria, and assembles this issue from the result.

The thematic profile is **aligned with the research interests of Markus L. Fischer** (Palaeoclimate Dynamics group, University of Potsdam): Quaternary palaeoclimate and palaeoenvironmental reconstruction, the East African Rift, the African Humid Period and human-environment coupling, lake-level and water-balance modelling, palaeofloods and fluvial geomorphology, ICDP drill cores, and linear and nonlinear time-series analysis. Anyone working in these fields will probably find something useful here — it is explicitly not a complete survey of Quaternary science.

Selection, scoring and summaries are produced by machine and reviewed before publication. Details may contain errors; where in doubt, the linked original source governs.

Each paper receives a score from 0 to 100 across five criteria: thematic fit (40), methodological fit (25), impact/venue (15), applicability (10) and recency (10). Categories: ● 75 and above, must-read · ● 50–74, relevant · ● 30–49, on the radar. The score measures proximity to this newsletter's profile and sets reading priority; it is not a quality judgement on the individual paper.

In brief

Five must-reads, and the week is unusually strong at the edges of the profile. From the core region comes a sediment-budget study of the **Omo catchment and its contribution to the Turkana Depression** across 27 million years — the same basin, a different timescale. Methodologically closest is a **QSR paper on Mojave palaeohydrology during the last deglaciation**, combining core sequences with shoreline archives; this is the reference architecture for robust lake-level reconstruction in arid basins. The most striking result: **Nature Geoscience** shows, from a 10,600-year Congo peat core, that dry phases cut the net oxygen budget by about 80 percent within centuries — abrupt, nonlinear, hydroclimatically triggered. Two palaeoflood papers follow, from the Yellow River and the Dadu Gorge.

On funding, the most important news of the year: the dedicated palaeoclimate call **HORIZON-CL5-2027-01-D1-08** now has firm dates — opening **17 November 2026**, deadline **4 March 2027**, around €14 million for roughly two projects. At that success rate, consortium building is decided in the coming weeks, not in November. Alongside it: the **ERC Starting Grant 2027 with a 15 October 2026 deadline** and **PAGES workshop applications due 11 September 2026**.

Politically uncomfortable: the DFG has **suspended approval of postdoc positions in Research Training Groups** and pushed the next Priority Programme round to late 2026, with funding starting only in 2028. The draft revision of the German fixed-term academic contract law (WissZeitVG) introduces minimum contract durations of two years for postdocs but removes the carry-over provision — on balance a deterioration for the postdoc phase.

Part A — Papers of the week

Papers were captured with a Crossref `created` date from 6 July 2026 onwards. Where Elsevier supplies no abstract to Crossref, the assessment rests on title, venue and thematic context — without inventing results. Such entries are marked `△ title-based`.

● Must-read (75 and above)

- 1. Paleohydrologic conditions in the Mojave Desert during the last deglaciation recorded by core-to-shore lacustrine records** · *Quaternary Science Reviews*, 2026 · **Score 93** Lake-level and water-balance reconstruction in an arid basin, built from the combination of drill core and shoreline or beach-ridge archives. This architecture — a core signal plus geomorphological evidence from the basin margin — is what allows lake levels to be anchored in absolute rather than merely relative terms, and it transfers directly to East African rift basins. The deglacial focus makes the paper additionally useful as a comparison case for termination dynamics. `△ Title-based (no abstract via Crossref/Elsevier)`. Theme 40 · Method 22 · Venue 11 · Applicability 10 · Recency 10 (created 18 Jul 2026) · DOI: [10.1016/j.quascirev.2026.110171](https://doi.org/10.1016/j.quascirev.2026.110171)
- 2. Hydroclimate controls Congo peatland net oxygen release over the past 10,600 years** · *Nature Geoscience*, 2026 · **Score 91** Direct redox titration applied to a 10,600-year peat core from the Congo Basin. The authors find systematic deviation from the conventionally assumed 1:1 oxygen-to-carbon stoichiometry and use it to quantify the net redox imbalance through time. The central finding is that the fluxes behave strongly nonlinearly: drier phases triggered abrupt declines in the oxygen surplus of about 80 percent within centuries, followed by rapid recovery once wetter conditions returned. Doubly relevant here — African Holocene hydroclimate on one hand, a well-documented case of threshold-like system response on the other. Theme 40 · Method 20 · Venue 15 · Applicability 10 · Recency 6 (created 10 Jul 2026) · DOI: [10.1038/s41561-026-02031-z](https://doi.org/10.1038/s41561-026-02031-z)
- 3. Palaeoflood Records from the Sanhuajian Reach in the Mid-lower Yellow River: Implications for Channel Evolution and Climatic Drivers** · *Palaeogeography, Palaeoclimatology, Palaeoecology*, 2026 · **Score 86** Palaeoflood reconstruction in a large river system, linked to channel evolution and climatic drivers. The methodological question is the same one that central European palaeoflood studies face, for instance on the Ahr: how to extrapolate robustly from slackwater and overbank deposits to discharge magnitudes, and how to separate the climatic signal from the geomorphological response of the system. `△ Title-based`. Theme 40 · Method 22 · Venue 8 · Applicability 10 · Recency 6 (created 6 Jul 2026) · DOI: [10.1016/j.palaeo.2026.114046](https://doi.org/10.1016/j.palaeo.2026.114046)
- 4. Mid-Holocene outburst floods from landslide-dam failures in the upper Dadu River Gorge, eastern Tibetan Plateau** · *Palaeogeography, Palaeoclimatology, Palaeoecology*, 2026 · **Score 84** Outburst floods from landslide dams during the mid-Holocene. Relevant as a contrasting case to climatically driven flood generation: such events produce extreme discharges without any climatic signal and are therefore a central source of error in interpreting palaeoflood series. `△ Title-based`. Theme 40 · Method 20 · Venue 8 · Applicability 6 · Recency 10 (created 17 Jul 2026) · DOI: [10.1016/j.palaeo.2026.114073](https://doi.org/10.1016/j.palaeo.2026.114073)
- 5. 27 million years of sediment production in the Omo River catchment and contribution to the sedimentation in the Turkana Depression (EARS, Kenya/Ethiopia)** · *Journal of African Earth Sciences*, 2026 · **Score 77** Sediment production in the Omo catchment

and its contribution to filling the Turkana Depression. The timescale reaches far beyond the Quaternary, but the paper supplies the long-term framework of sediment source areas and accumulation rates within which the region's Quaternary core archives sit — useful for contextualising sedimentation rates in Turkana and Chew Bahir cores. Δ Title-based. Theme 40 · Method 15 · Venue 6 · Applicability 10 · Recency 6 (created 9 Jul 2026) · DOI: [10.1016/j.jafrearsci.2026.106271](https://doi.org/10.1016/j.jafrearsci.2026.106271)

● Relevant (50–74)

6. Climate-induced forest destabilization and shrub stabilization in Africa · *Global and Planetary Change*, 2026 · **Score 72** Destabilisation of forest and stabilisation of shrub systems under climate forcing in Africa. Conceptually and methodologically close to work on resilience loss and early-warning signals, and therefore potentially connected to research on the termination of the African Humid Period. Without an abstract it remains open whether stability metrics or merely trend analyses were used. Δ Title-based; method assignment uncertain. Theme 25 · Method 18 · Venue 9 · Applicability 10 · Recency 10 (created 16 Jul 2026) · DOI: [10.1016/j.gloplacha.2026.105627](https://doi.org/10.1016/j.gloplacha.2026.105627)

7. Late-Holocene sediment dynamics and contaminants in the Fecht River, Vosges Mountains, France: From medieval mining to modern floodplains · *CATENA*, 2026 · **Score 70** Late Holocene floodplain sediment dynamics in a central European upland river, using heavy metals from medieval mining as chronomarkers. Regionally and methodologically close to central European floodplain and palaeoflood studies; the contamination approach transfers as an independent stratigraphic anchor for floodplain profiles. Δ Title-based. Theme 25 · Method 18 · Venue 7 · Applicability 10 · Recency 10 (created 15 Jul 2026) · DOI: [10.1016/j.catena.2026.110417](https://doi.org/10.1016/j.catena.2026.110417)

8. Millennial- to multi-centennial-scale Asian summer monsoon variability during Termination I: Evidence from stalagmite $\delta^{18}\text{O}$ in Wanxiang Cave, northwestern China · *Quaternary Science Reviews*, 2026 · **Score 66** A high-resolution speleothem time series across Termination I. Interesting as a comparison archive for termination dynamics: whether monsoon systems switch gradually or in jumps is the same question addressed for the AHP termination using early-warning signals. Δ Title-based. Theme 25 · Method 18 · Venue 11 · Applicability 6 · Recency 6 (created 10 Jul 2026) · DOI: [10.1016/j.quascirev.2026.110168](https://doi.org/10.1016/j.quascirev.2026.110168)

9. High-resolution imaging of a sediment core reconstructs past monsoon seasonality · *Nature Geoscience*, 2026 · **Score 65** — *News & Views, not a primary paper* A commentary on a study that uses high-resolution geochemical imaging of an Arabian Sea core to extract seasonally resolved signals, showing that the South Asian monsoon responded differently by season to global and high-latitude forcing during the last deglaciation. Methodologically relevant as a consistent extension of μXRF core scanning towards sub-annual resolution — an approach of general interest for laminated core intervals. Theme 25 · Method 20 · Venue 8 · Applicability 6 · Recency 6 (created 14 Jul 2026) · DOI: [10.1038/s41561-026-02025-x](https://doi.org/10.1038/s41561-026-02025-x)

10. Biogeochemical, sedimentological and dinoflagellate-based reconstruction of the Holocene palaeoenvironment of the east Limfjord, Denmark · *Boreas*, 2026 · **Score 64** A multiproxy reconstruction of a succession spanning roughly 9600 to 150 cal. a BP, with an AMS ^{14}C chronology built on terrestrial macrofossils. The paper dates the marine transgression at Sebbesund to about 9400–9300 cal. a BP and shows that salinity changes largely follow regional sea-level change. Worth reading above all as a clean example of coupling biogeochemistry,

● On the radar

- **Adaptive responses of prehistoric humans in the Nihewan Basin to Heinrich events during the last glacial period** · *QSR* · **63** ● · Human–environment coupling across abrupt climate events; a conceptual analogue to the African Humid Period question in a different region. DOI: [10.1016/j.quascirev.2026.110163](https://doi.org/10.1016/j.quascirev.2026.110163)
- **Limnological responses to fire and carbonate dynamics in a tropical shallow lake (central Mexico)** · *Quaternary International* · **62** ● · Diatoms, titanium and macrocharcoal combined; a transferable multiproxy approach for shallow tropical lakes. DOI: [10.1016/j.quaint.2026.110392](https://doi.org/10.1016/j.quaint.2026.110392)
- **Diachronic change in dietary behavior of Plio-Pleistocene *Theropithecus* from South Africa** · *P³* · **62** ● · Palaeoecology in the hominin context, relevant to environmental reconstruction at African fossil sites. DOI: [10.1016/j.palaeo.2026.114063](https://doi.org/10.1016/j.palaeo.2026.114063)
- **Late Pleistocene–Holocene pollen, macrofossils, and diatoms from eastern New York state's Rensselaer Plateau** · *Quaternary Research* · **60** ● · A multiproxy sequence since retreat of the Laurentide ice sheet, with several temperature and hydrological shifts. DOI: [10.1017/qua.2026.10096](https://doi.org/10.1017/qua.2026.10096)
- **Making palaeoclimatology research policy-relevant: Insights and experience from the DECADAL Project** · *JQS* · **53** ● · An experience report on translating palaeoclimate findings into policy advice; useful for impact sections in proposals. DOI: [10.1002/jqs.70096](https://doi.org/10.1002/jqs.70096)
- **Hydroclimate variability in Northeast China during Marine Isotope Stage 3** · *Global and Planetary Change* · **48** ● · Hydroclimate reconstruction in MIS 3, thematically peripheral. DOI: [10.1016/j.gloplacha.2026.105623](https://doi.org/10.1016/j.gloplacha.2026.105623)
- **Holocene glacial activity on Svalbard and its impact on iron delivery** · *QSR* · **45** ● · Glacial–marine coupling and the proxy interpretation of iron delivery. DOI: [10.1016/j.quascirev.2026.110183](https://doi.org/10.1016/j.quascirev.2026.110183)

Part B — Funding & calls

Horizon Europe — "Palaeoclimate science for a better understanding of Earth system dynamics" (HORIZON-CL5-2027-01-D1-08). Opening **17 November 2026**, deadline **4 March 2027**, Research and Innovation Action, budget around **€14 million for roughly two projects**, so about €7 million per consortium. The stated aim is robust information on palaeoclimate states and events outside the range of variability recorded over the past centuries, to guide the development of Earth system models. This is the tailor-made call previously only announced in the work programme; with two funded projects Europe-wide the competition is severe, and consortium discussions belong early in the timeline. Sources: [topic page](#) · [CL5 Work Programme 2026–2027 \(PDF\)](#).

ERC Starting Grant 2027 — deadline 15 October 2026. Step 1 results in April 2027, interviews and Step 2 results in August 2027. Eligibility: PhD defended no more than ten years before 1 January 2027, extendable for parental and care leave. New from the 2027 calls: at most one Starting and one Consolidator Grant per career, plus stricter application rules. The **ERC**

Consolidator 2027 is expected in January 2027. Sources: [ERC — Applying in the 2027 competitions](#) · [ERC — Changes to the 2026 and 2027 Work Programmes](#) · [Science|Business](#).

PAGES — working group and workshop support, deadline 11 September 2026. The next application round covers workshops, meetings, new working groups and extensions of existing groups. Ahead of the Horizon palaeoclimate call this is a convenient instrument for forming a visible consortium. Also: the **PAGES Early-Career Award 2027**, applications close **29 September 2026**. Sources: [PAGES — Workshop/meeting support](#) · [PAGES Newsletter July 2026](#).

ICDP — proposal cycle. The regular annual deadline for full, pre- and workshop proposals is **15 January**; the 2027 cycle is likely to follow the same pattern, though no formal announcement has appeared. Workshop calls run through the year. Source: [ICDP — Proposals](#).

DFG — changed conditions. No new palaeoclimate-specific call. Relevant for planning: the **programme allowance rises from 22 to 25 percent from 2027**; the next **Priority Programme application round** takes place only in **late 2026**, with funding starting in **2028**. Sources: [Forschung & Lehre](#) · [Fakultätentag Informatik](#).

Part C — Research policy (Germany/EU)

DFG cuts at the postdoc level. Facing high collective-agreement pay rises and increased equipment and consumables costs that the guaranteed annual funding increments do not cover, the DFG has postponed the funding start for new Research Training Group applications and **suspended approval of postdoc positions**. For postdoc career paths in the geosciences this removes one funding route for the time being, increasing pressure on individual grants and EU money. Source: [Fakultätentag Informatik](#).

Federal budget 2027 — cabinet decision expected 29 July 2026. The research ministry's budget remains nearly stable against the April planning, around €70 million below it, making it an exception compared with other departmental budgets; growth in the federal budget goes largely to defence. As interpretation rather than fact: stability at a high nominal level amounts, given current pay and materials cost increases, to real stagnation — consistent with what the DFG reports operationally. Source: [Wiarda blog, 7 July 2026](#).

WissZeitVG — ministry draft published. The core of the draft is **minimum contract durations of three years for first contracts of doctoral researchers and two years for postdocs** in the respective qualification phase. The maximum fixed-term duration remains six years per phase; the "four plus two" arrangement from the failed previous draft is off the table. Unfavourable for the postdoc phase: the previous carry-over provision is removed, under which unused fixed-term time from the doctoral phase could be transferred. Assessments diverge widely, from "timid minimal compromise" to criticism from the university side. Practical consequence: anyone carrying unused time from the doctoral phase should check how the draft affects their personal fixed-term account. Sources: [Wiarda blog](#) · [Forschung & Lehre](#) · [fzs statement](#) · [DPG statement](#).

DFG annual meeting, 1 July 2026. A clear statement in defence of the freedom and independence of science, combined with criticism of the AfD's science policy. In parallel, five new Research Units were established with around €27 million in total, and five were extended. Source: [DFG](#).

Part D — Community & deadlines

- **ERC Starting Grant 2027** — deadline **15 October 2026**.
 - **PAGES working group and workshop applications** — **11 September 2026** · **PAGES Early-Career Award 2027** — **29 September 2026**.
 - **AGU Fall Meeting** — abstract deadline **5 August 2026** (per PAGES Newsletter July 2026).
 - **DEUQUA conference 2026** — **28 September to 2 October 2026 in Freiburg im Breisgau**, with full-day excursions on 28 September and 2 October. The Göttingen meeting originally planned for 2026 has been postponed to 2028. No abstract deadline was findable in the sources searched. Source: [DEUQUA](#).
 - **Horizon CL5 palaeoclimate call** — opening **17 November 2026**, deadline **4 March 2027**.
 - **EGU General Assembly 2027** — **4 to 9 April 2027 in Vienna**, hybrid. The abstract deadline is not yet published; experience suggests mid-January, and for EGU26 it was 15 January 2026. Source: [EGU](#).
 - **ICDP** — next regular proposal deadline expected **15 January 2027**, following previous years' pattern and not yet confirmed.
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Method & sources

Paper screening uses the Crossref REST API with topic- and ISSN-based queries and the `from-created-date` filter. Abstracts, where available, come from Crossref; Elsevier generally does not supply them, and those entries are marked $\triangle$ *title-based*. DOIs already listed in an earlier issue are not repeated. Funding, research policy and deadlines rest on the sources linked in Parts B to D.

Limitation of this issue: direct API access from the compute environment was blocked; the Crossref queries ran through the browser instead — the same source, but with a reduced number of results per query. Coverage is therefore somewhat lower than usual and very recent registrations may be missing. ScienceDirect additionally blocked retrieval of publisher abstracts, which raises the share of title-based assessments above the normal level.

GeoNews is an automatically generated digest published by Dr Markus L. Fischer, Palaeoclimate Dynamics group, University of Potsdam. The thematic profile follows his research interests; selection is made by a rule-based workflow, not an editorial team. The digest is an independent literature review with no affiliation to the publishers or funding bodies mentioned, and does not constitute a recommendation or advice. Corrections and pointers are welcome.